

MASENO 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

231/3

BIOLOGY

PAPER 3 (PRACTICAL)

TIME: 1¾ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- Write your Name, Class and Adm No. in the spaces provided above
- Answer ALL the questions in the spaces provided
- You are required to spend the first **15 minutes** of the **1 ¾ hours** reading the whole paper carefully before commencing your work.
- Candidates should answer the questions in English

FOR EXAMINERS USE ONLY

Question	Maximum score	Candidate's score
1	16	
2	11	
3	13	
Total Score	40.	

1. You are provided with a suspension labelled M.

a) Using the reagents provided only, test for the food substances present in the suspension. In the table below, record the food tested, procedure, observation and conclusions. **12 marks.**

Food TEST	Procedure	Observation	Conclusion

b) Name two enzymes that may be required to digest suspension M in the alimentary canal in human.

(2mark)

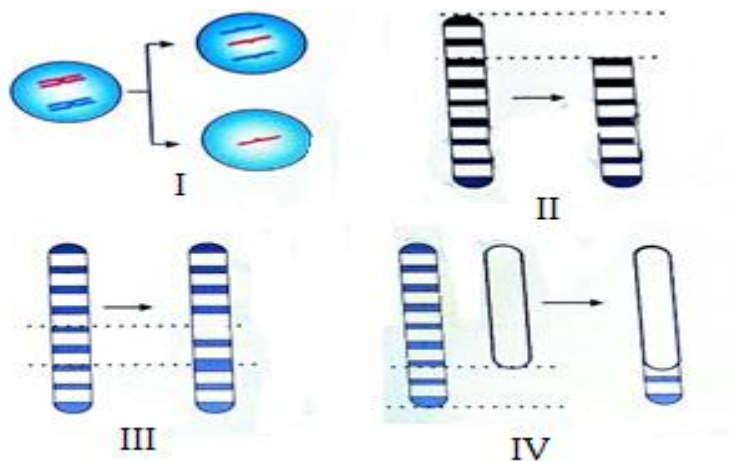
c) State role of the following in the experiment.

(2marks)

(i) Sodium hydrogen carbonate

(ii) Dilute hydrochloric acid

2. Experimental evidence shows that most mutations results to variations among organisms. The illustrations I, II, III and IV below show different possible types of chromosome mutations. Study them keenly.



a) **Identify** the types of chromosome mutations illustrated.

(4 marks)

Illustration	Name of mutation
I	
II	
III	
IV	

b) **State** the effect of the mutation illustrated in III.

(1 mark)

c) **In most** cases, the mutation in II could be lethal. Explain.

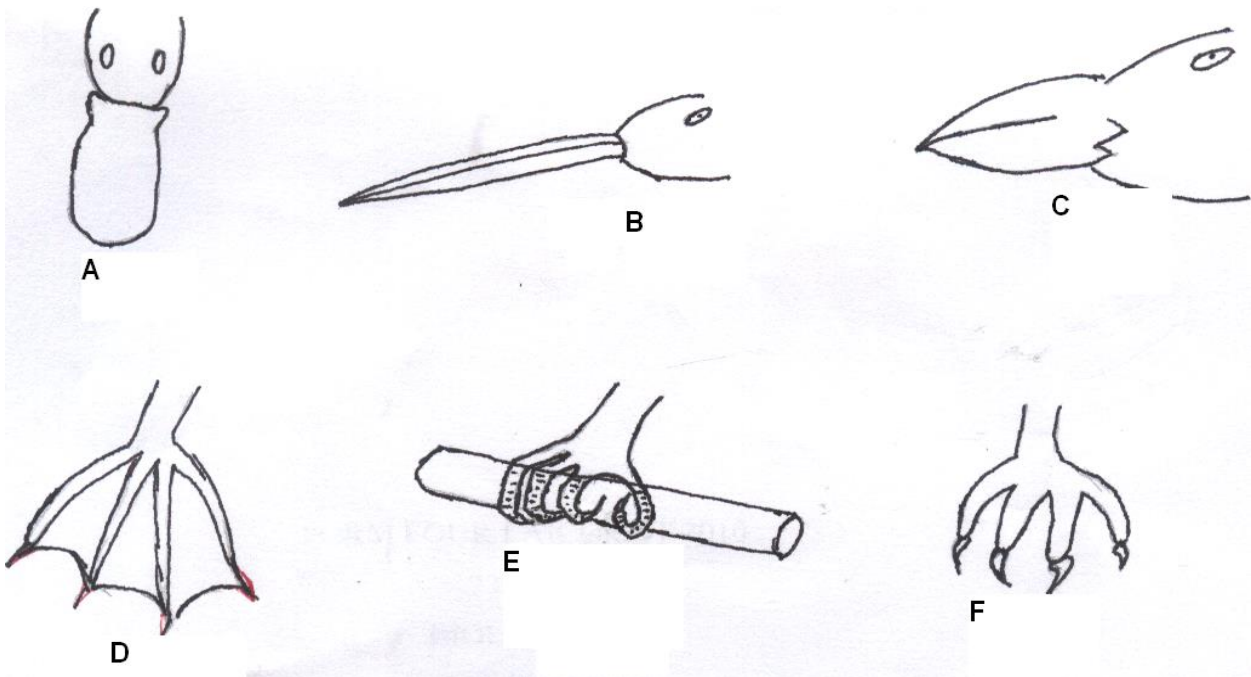
(2 marks)

d) During which stage of mitosis does the mutation in I above occur? **(1 mark)**

e) i) Define a mutation **(1 mark)**

ii) Give two possible environmental mutagens. **(2marks)**

3. The diagrams below represent body parts of some organisms (animals). Study them and answer the question that follow



a. (i) Suggest the type of food eaten by organisms with the parts labeled A, B, C and F (4mks)

Food

Reason

A

B

C

F

(ii) With reasons, suggest the likely habitat of the organism from which the parts labeled D and E were obtained

(4mks)

Part

Habitat

Reason

D

E

b. (i) Suggest the type of evolution that is exemplified by the organisms labeled D, E and F. Give reason for your answer.

The type of **evolution**

(2mks)

Reason

(ii) Suggest the significance of the above named type of evolution for the organism **(1mk)**

c. (i) Explain briefly the meaning of analogous **structures** **(1mk)**

(ii) Give one example of analogous structure **(1mk)**